Day 7 Task — Linux Terminal Lab: Answer Guide

Tekloomy Cyber Security Internship · Module 2: Know Your Battlefield · Day 7

How to use: run each command in your Ubuntu VM and replace the bracketed/bold items (counts, PIDs, ports) with what *your* VM shows. A screenshot of the terminal for each task is the fastest way to submit, and the task sheet says it is welcome. Take a VM snapshot first.

Step 1 — The 20 tasks

#	Area	Task	Command(s) to use	Result / note
1	Navigation	Current dir + all files, long format	`pwd` `ls -la ~`	Prints /home/<you>. Long list incl. hidden files (.bashrc, .profile, .ssh…).
2	Navigation	/var/log, 5 most recent files	`cd /var/log` `ls -lt | head -6`	Newest first; head -6 because the first line is “total”. Typical: syslog, auth.log, kern.log.
3	Files	Create ~/seclab/notes and scripts	`mkdir -p ~/seclab/{notes,scripts}`	No output; -p creates parent and both children in one command.

4	Files	day7.txt with the line	`cd ~/seclab/notes`
			`echo "Linux lab started" > day7.txt`
			`cat day7.txt`
				cat shows: Linux lab started

5	Files	Copy then rename	`cp day7.txt backup.txt`
			`mv backup.txt day7-backup.txt`
			`ls`
				ls shows day7.txt and day7-backup.txt.

6	Search	.conf files under /etc, count	`find /etc -name "*.conf" 2>/dev/null | wc -l`	A number (roughly 100–200 on a fresh VM): **write yours**. 2>/dev/null hides permission errors.
7	Search	Lines with 'bash' in /etc/passwd	`grep bash /etc/passwd` `grep -c bash /etc/passwd`	Usually 2 real login users: root and your own user (others use nologin/false). **Write your count.**
8	Users	Username and groups	`whoami` `groups`	Shows your name and groups. Yes, sudo is in the list (first user on Ubuntu is added to sudo).
9	Users	Create user analyst with home	`sudo useradd -m analyst` `(or: sudo adduser analyst)`	-m creates /home/analyst. Check: ls /home

10	Users	Set password, switch, whoami, exit	`sudo passwd analyst`
			`su - analyst`
			`whoami`
			`exit`
				whoami prints: analyst. exit returns to your user. (Do not write the password on the sheet.)

11	Permissions	day7.txt owner-only	`chmod 600 ~/seclab/notes/day7.txt` `ls -l ~/seclab/notes/day7.txt`	Shows -rw------- (owner read/write only).

12	Permissions	hello.sh executable, run	`cd ~/seclab/scripts`
			`echo 'echo Hello from $(whoami)' > hello.sh`
			`chmod +x hello.sh`
			`./hello.sh`
				Prints: Hello from <your username>. Single quotes stop $(whoami) from running early.

#	Area	Task	Command(s) to use	Result / note
13	Permissions	Find SUID binaries, count	`find / -perm -4000 -type f 2>/dev/null | wc -l` `find / -perm -4000 -type f 2>/dev/null`	Count: **write yours** (roughly 20–40). Recognisable: /usr/bin/passwd, /usr/bin/sudo, /usr/bin/su (also mount, chsh, gpasswd).
14	Processes	List processes, find shell PID	`ps aux | grep bash` `echo $$`	echo $$ prints your shell's PID (a number).

15	Processes	Background sleep, find PID, kill	`sleep 300 &`
			`pgrep sleep`
			`kill <PID>`
			`pgrep sleep`
				First pgrep shows the PID; after kill the second shows nothing, so it is gone. (jobs also works.)

16	Logs	Last 20 lines of auth.log	`sudo tail -n 20 /var/log/auth.log` `(or: sudo journalctl -n 20)`	Login sessions, sudo commands, su to analyst, user creation/password change from earlier tasks.
17	Logs	Count 'sudo' in auth.log	`sudo grep -c sudo /var/log/auth.log` `(or: journalctl | grep -c sudo)`	A number: **write yours**. It is high because each of your sudo commands is logged.
18	Network	Listening TCP/UDP ports + process	`sudo ss -tulpn` `(or: sudo netstat -tulpn)`	Typical: 127.0.0.53:53 (systemd-resolved), UDP 68 (DHCP client), maybe 631 (CUPS) and 5353 (avahi). **Write yours.**

19	Packages	Update lists, install tree	`sudo apt update`
			`sudo apt install tree -y`
			`tree ~/seclab`
				Tree shows seclab → notes (day7.txt, day7-backup.txt) and scripts (hello.sh).

20	Scripting	check.sh: date, user, logged-in users	`cd ~/seclab/scripts`
			`cat > check.sh << 'EOF'`
			`#!/bin/bash`
			`date`
			`whoami`
			`who | wc -l`
			`EOF`
			`chmod +x check.sh`
			`./check.sh`
				Prints today's date, your username and a number (e.g. 1).

Step 2 — Investigate a suspicious server

Step	Command	What would worry you in the output?
Who	`last -a (and: who, w)`	Logins from unknown users or odd IP addresses, logins at strange hours, accounts that should not exist, extra users with UID 0.
What	`ps aux --sort=-%cpu | head`	Unknown or oddly named processes, very high CPU/RAM (cryptominer), processes running from /tmp or /dev/shm, processes owned by www-data spawning shells.
Where	`sudo ss -tulpn`	Unexpected listening ports (e.g. 4444, 6667, high random ports), established connections to unknown foreign IPs, a process listening that you do not recognise.
When	`sudo tail -n 100 /var/log/auth.log (or journalctl)`	Many “Failed password” lines then an “Accepted”, which means brute force succeeded; new users added; sudo use at odd times. This gives the timeline of the attack.

Step	Command	What would worry you in the output?
How	`sudo find / -mtime -2 -type f 2>/dev/null | head (also: crontab -l; cat ~/.ssh/authorized_keys)`	Recently changed files in /tmp, /var/www (web shell), /etc; unknown cron jobs; unfamiliar SSH keys. This shows how the attacker got in and stays in.
Fix	`sudo kill <PID>; sudo passwd -l <baduser>; sudo apt update && sudo apt upgrade -y; sudo ufw enable`	Confirm the malicious process is gone and the account locked. Patch the hole, rotate passwords/keys, enable the firewall, and restore from a clean backup if needed.

Step 3 — Get ready for Day 8

```
sudo apt install wireshark # answer Yes to “non-superusers should be able to capture packets”
sudo usermod -aG wireshark $USER
sudo reboot
```

After the reboot, run `groups` and check that **wireshark** is in the list. Then tick: [ ] done.